

This appendix is supplementary to [MANU-SCRIPT_v12_EN.md](#). Its contents are not used directly in the main argument but are provided separately so that the symbol definitions, model validation, and sensitiv-

ity to assumptions can be reproduced and checked. To be attached as supplementary material at submission.

Appendix — Symbol System, Model Validation, and Sensitivity to Assumptions

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manuscript, and the companion tool [design_simulator.html](#), share these definitions.

A-1. Entity and symbol system

All entities and symbols used in this study are defined uniformly in Table A1. Every equation and table in the

Table A1. Definition of entities and symbols

Symbol	Entity	Definition / unit	Character
a	age band	five-year bands (20–24 ... 85+) or age group g	index
h	household size	1, 2, 3, 4, 5+ persons	index
s	area band	small <60m ² / standard 60–85m ² / large ≥85m ²	index
r	room band	1–2 rooms / 3 rooms / 4+ rooms	index
k	tenure form	sale / jeonse / monthly rent	index
t	year	2013 ... 2035	index
n _a	net inflow	net increase for age band a (persons)	measured (DT_1B04005N differences)
ρ _a	headship	household-head rate by age (dimensionless)	measured (DT_1JC1511 ÷ population)
P(h a)	conditional household-size distribution	household-size distribution for head age a	measured (DT_1JC1511)
$\bar{m}$	mean household size	observed Godeok anchor (persons per household)	measured band 2.99–3.46
q _h , p _h	household-size distribution	before / after correction	derived
M ^A _(h,s) , M ^R _(h,r)	size and room mapping	household size → area and room propensity	assumed (sensitivity-tested)
T _(s,k)	tenure propensity	sale / jeonse / rent shares by area	measured (turnover-adjusted to stock)
D ^A _s , D ^R _r , D ^K _k	demand	demand share (%) and units by area, rooms, tenure	derived
S _s	supply stock	composition of existing supply (units, %)	measured (complex metadata)
G _s	gap	S _s – D ^A _s (negative = shortage, positive = surplus)	derived
K _{pop}	planned carrying population	144,173 persons (ceiling clip in the forecast)	published (development plan)

Source: KOSIS, MOLIT real transactions, NAVER complex metadata, Godeok International New Town development plan.

A-2. Sensitivity to the unit-type mapping assumption

As stated in §III.3-1) of the manuscript, among the inputs to the chain, headship ρ_a, the conditional distribution P(h|a), and tenure propensity T_(s,k) are

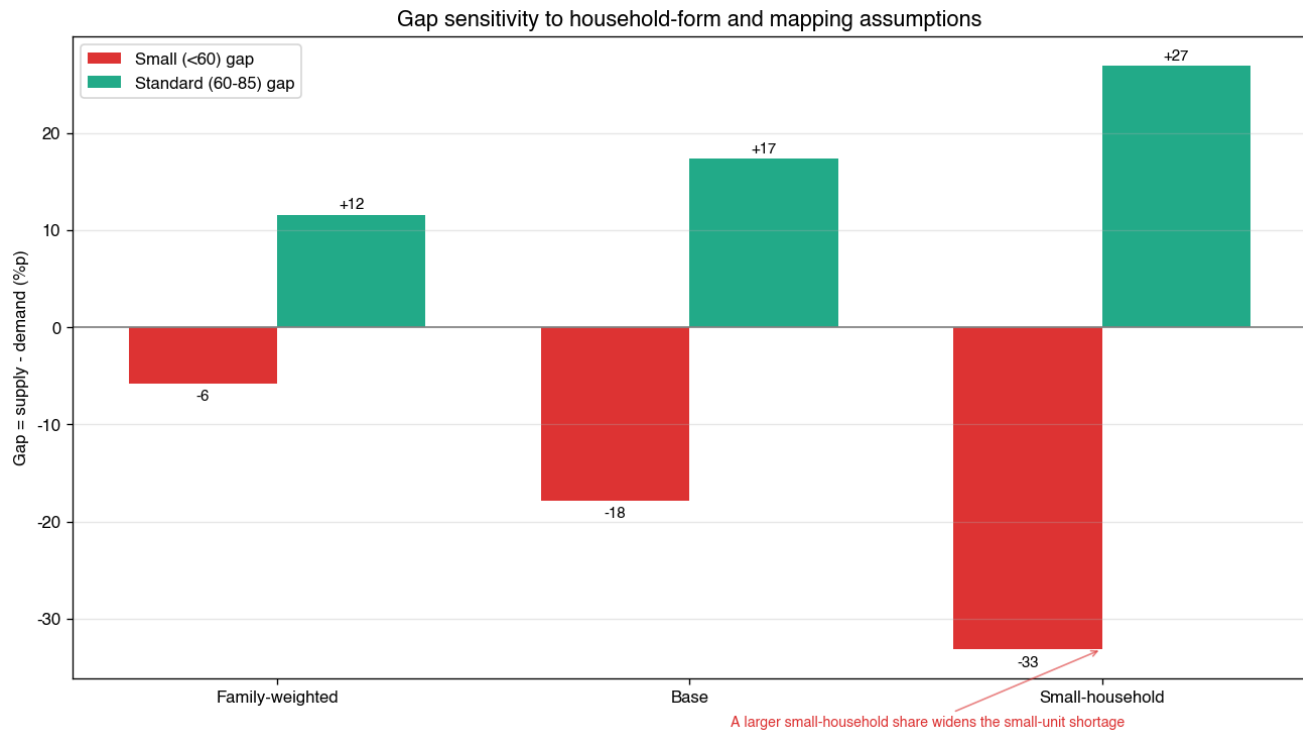
measured, and **only the unit-type mapping M is assumed**. Whether varying M flips the sign of the gap therefore governs the robustness of the conclusions.

Varying M across three scenarios (strong small-unit preference, baseline, strong large-unit preference), **the 1–2-room shortage on the room axis retains its sign in every scenario** (–15.7 to –25.4pp). The floor-

area axis, by contrast, is sensitive to the mean-household-size assumption $\bar{m}$: under the first pass ($\bar{m}=2.18$) small units are short by 5.8 to 33.2pp, whereas under the Godeok-specific re-estimate ($\bar{m}=2.99-3.46$) they are in surplus by 0.6 to 9.5pp — a sign reversal.

This is the basis for the statement in §V.2 that the room-axis conclusion is robust to parameter choice while the area-axis conclusion is conditional.

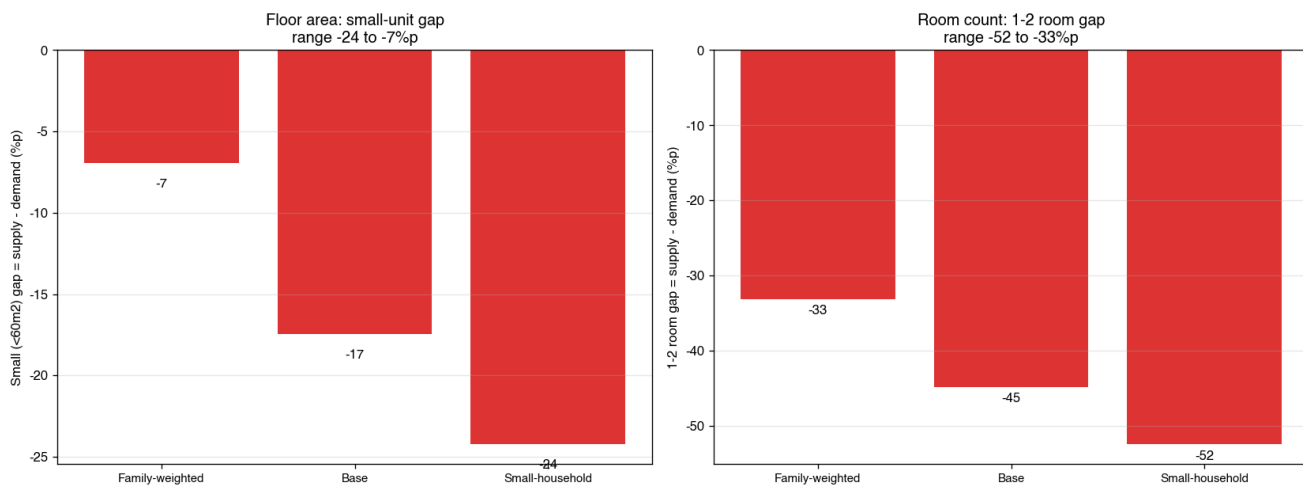
Figure A1. Sensitivity of the gap to mapping assumptions



Source: author's calculation (`analysis/21b\_gap\_sensitivity.py`). This figure is on a first-pass basis ($\bar{m}=2.18$), so its indication of a small-unit shortage pertains to the parameter set rejected in §IV.4-(1). It should be read only for **the relative magnitude of sensitivity** to mapping variation.

Figure A3. Model sensitivity of the design chain

Sensitivity: the direction of the gap is robust to mapping assumptions



Source: author's calculation (`analysis/24b\_model\_sensitivity.py`).

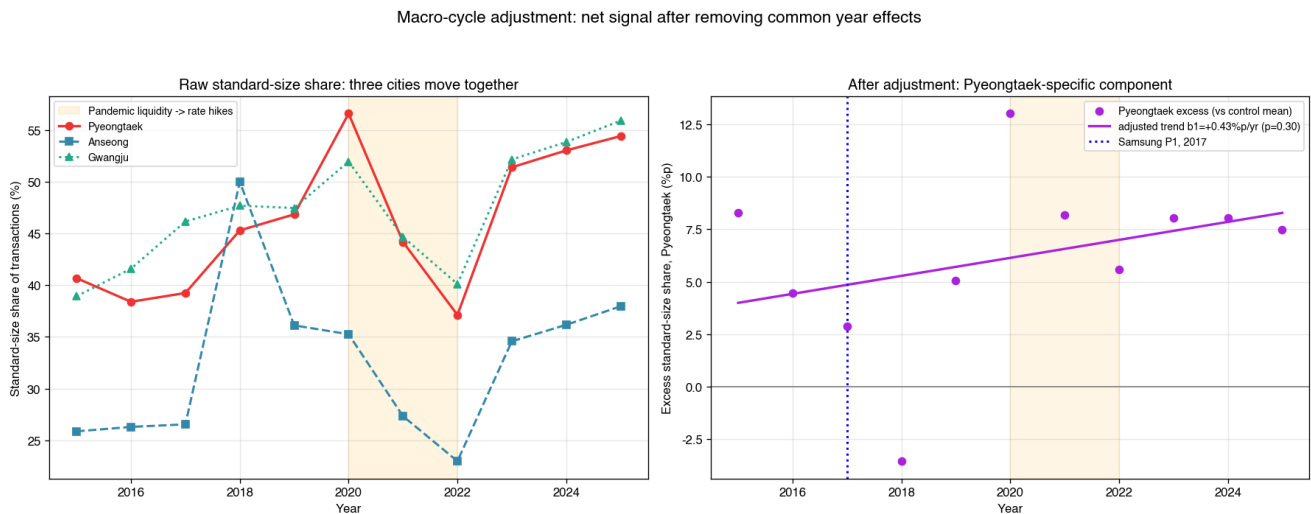
A-3. The nationwide character of the unit-size trend — background check

This is the evidence behind the background check summarized in §IV.1 of the manuscript. Convergence on the standard size and the decline of small units appear

equally or more strongly in Anseong and Gwangju (before any semiconductor investment), and adjusting for macro cycles (pandemic, interest rates) removes any Pyeongtaek-specific trend. This study therefore does not attribute the unit-size *trend* itself to a semiconductor effect; its focus is the **consistency between Godeok's**

local population structure and its supply composition.

Figure A2. Nationwide nature of the unit-size trend — macro-cycle adjusted



A-4. Design model — before/after correction and the tenure turnover adjustment

fore and after recalibration (Gap = supply – demand; negative = shortage)

Table A4. Design-model results for Godeok, be-

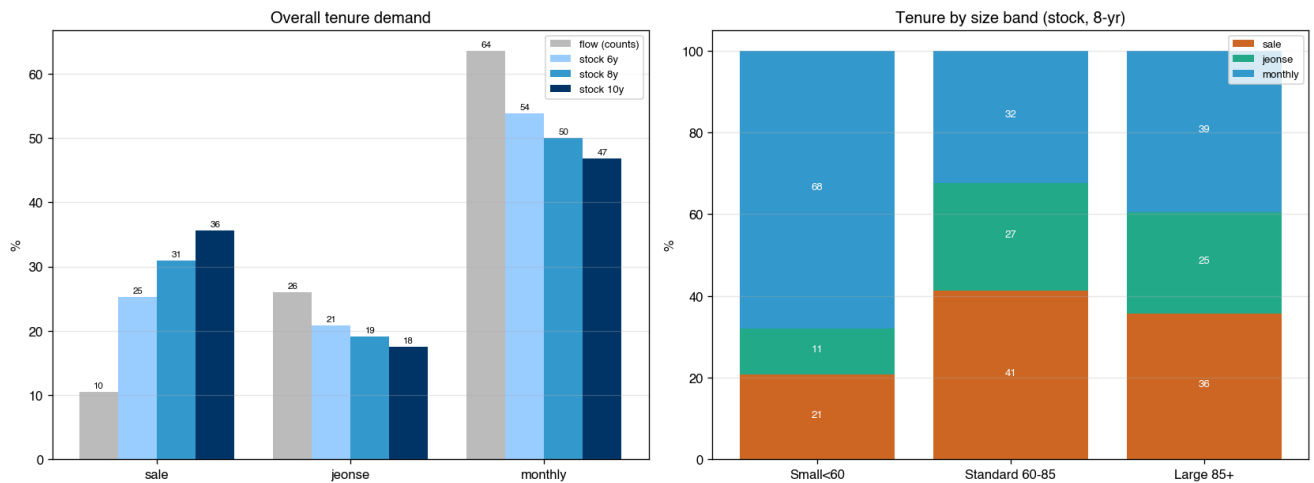
Indicator	Before (city-wide parameters)	After (Godeok-specific, baseline $\bar{m}=3.23$)
Single-person share of new inflow	39.3%	16.1%
Mean household size of the distribution	2.22 (1.2 persons from observation)	3.23 (consistent with observation)
Small-unit area gap	-17.4pp shortage	+5.2pp surplus
Large-unit area gap	+0.2pp	-12.5pp shortage
1-2-room gap	-44.8pp	-20.4pp shortage (direction held)
Tenure (flow→stock adjusted)	sale 31 / jeonse 19 / rent 50%	sale 35 / jeonse 22 / rent 43%

Source: author's calculation (`analysis/24\_design\_model.py`, `26\_tenure\_stock\_adjust.py`, `32\_godeok\_recalibration.py`, `36\_forecast\_design\_recalibrated.py`). The "before" column pertains to the **rejected parameter set** and is presented only for contrast with the re-estimate.

Two adjustments compound on the tenure axis. **First, the flow→stock adjustment** is required because transactions are **counts**, so fast-turnover monthly rentals are over-sampled (raw 64% → 50% stock-adjusted). **Second, the Godeok-specific correction** reduces the share of small households, shrinking small-unit demand and with it the rent propensity concentrated in

small units, bringing monthly-rent demand down once more from 50% to **43%**. Even with both adjustments, monthly-rent demand remains in the low 40s — locally high relative to control cities, and not enough to overturn the unit-diversification or access conclusions.

Figure A4. Tenure turnover adjustment — flow vs. stock



Source: author's calculation (`analysis/26\_tenure\_stock\_adjust.py`).

A-5. Model validation — cross-city transfer and time-axis backcast

Table A2. Cross-city prediction error for household size (MAPE)

City	MAPE (%)	Interpretation
Pyeongtaek (self-consistency)	0.8	chain mechanically accurate
Anseong (similar control)	6.0	good transfer
Hwaseong (family-heavy new town)	11.0	small households over-predicted (local difference)

Source: author's calculation (`analysis/25\_model\_validation.py`).

Table A3. Rolling-origin backcast comparison of population forecast models

Model	Mean MAPE (%)	Verdict
Logistic (K fixed)	33.8	rejected — consistent over-prediction
Linear (recent three-year slope)	5.6	adopted
Damped linear ($\phi=0.8$)	13.3	runner-up

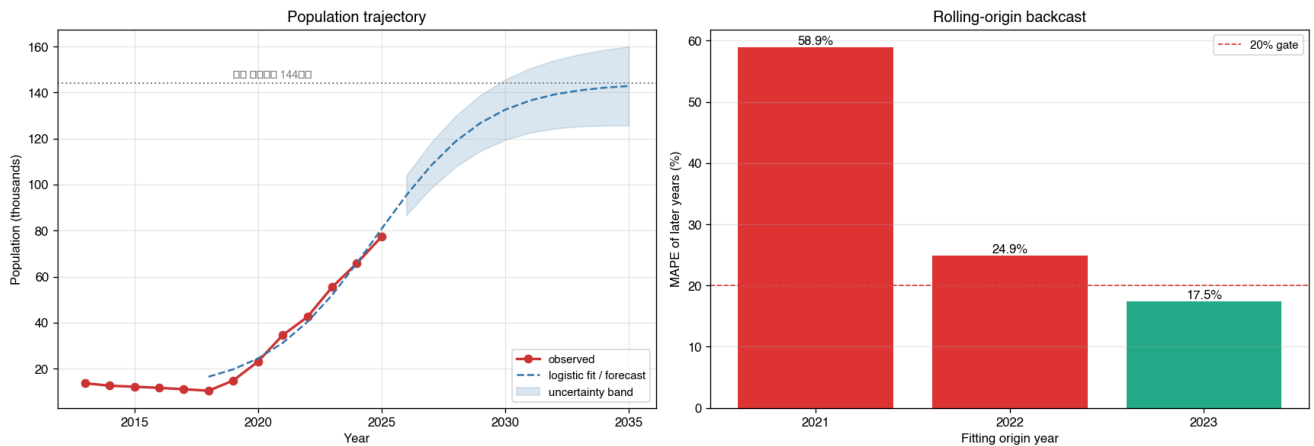
Source: author's calculation (`analysis/31\_backcast\_validation.py`). Mean of prediction errors for subsequent years from origins 2021, 2022, and 2023 respectively.

The methodological significance of rejecting the logistic model matches the first lesson in §V.2. Pinning the saturation point to the planned carrying population extrapolates the slope of the early rapid-growth phase as a

saturation rate, and **that bias did not surface without ex post validation (backcasting).**

Figure A5. Backcast validation of the population forecast

Population trajectory — backcast MAPE 33.8%



Source: author's calculation (`analysis/31\_backcast\_validation.py`).

A-6. Inflow-profile sensitivity in the companion tool

The companion tool [design_simulator.html](#) referenced in §IV.7-(6) offers two modes for mean household size: ① an **auto-linked mode** that estimates it from age composition via headship and corrects for the meas-

ured Godeok bias (observed 3.23 ÷ estimated 2.48 = ×1.30), and ② a **direct-entry mode** for sensitivity checks. In auto-linked mode, changing the inflow profile moves the results as in Table A5.

Table A5. Results by inflow profile (simulator, auto-linked mode)

Inflow profile	Mean household size	Single-person share	Small-unit gap	Large-unit gap	1-2-room gap
Godeok as measured	3.23	16.2%	+5.2	-12.6	-20.3
Young-worker concentrated	3.01	21.2%	+0.6	-9.9	-25.3
Family-centered	3.57	10.7%	+11.4	-17.0	-13.6

Source: companion tool output (`design\_simulator.html`). Developers can substitute their own demand projections to reproduce or revise the mix presented in this study.

Across all three profiles the **1-2-room and large-unit shortages retain their sign** and small units remain in surplus. Lowering mean household size to 2.18 (Py-eongtaek as a whole) in direct-entry mode reproduces the first pass's "small units short" conclusion, making visible how parameter choice changes the conclusion.

A-7. Figures excluded from the manuscript (for reproduction)

The following figures were excluded from the manuscript because they are not used directly in its argumentative chain or duplicate another exhibit, but **the PNG files and generating scripts remain under analysis/** and can be consulted for reproduction or extension.

File	Content	Reason for exclusion
28_supply_structure.png	supply distribution by rooms and area	Table 2 carries the same figures, so the chart adds no information
32_godeok_recalibration.png	three-panel recalibration	absorbed into consolidated Figure 3-①-③
34_barbell_distribution.png	three-panel distributional mismatch	absorbed into consolidated Figure 3-②-④
13_development_concentration.png	spatial concentration of development (Godeok share 1→73%)	same message as Figure 1
12_local_premium_gradient.png	local price gradient by distance	subset of Figure 5 (continuous-distance hedonic)
21_gap_index_prototype.png	first-pass gap index	visualization of the parameter set rejected in §IV.4-(1) — retaining it invites misreading

File	Content	Reason for exclusion
16_hedonic.png	premium controlling for building age	redundant with Figure 5
24_design_model.png	schematic of the design chain	redundant with the equations in §III.3-3) and Table A4
25_model_validation.png	cross-city validation	Table A2 (three rows) suffices
27_forecast_design.png	design-target trajectory	merged into Figure 6 (annual roadmap). based on uncorrected parameters (§V.4-(5))

Unsubmitted student manuscript (draft) – Appendix